MANIPAL UNIVERSITY JAIPUR

Statistical Methods and Probability Theory (MAS2001)

Session: Jan-May 2025

Assignment #1

(Probability Theory and Random Variables: Probability (Only One Lecture), Random variables, Cumulative distribution functions, Discrete random variables, Continuous random variables, Independent random variables, Probability mass and density functions, Expectation of random variables, Chebyschev's inequality.)

SECTION A (Memory Based Questions)

Q1.	(a) Define random variable.
	(b) If two random variables X and Y are independent, then ……..
	(c) Write the statement of Chebyschev's inequality.
	(d) Define distribution function.
	(e) Is the function defined as follows a density function? $f(x) = \begin{cases} e^{-x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$
	(f) A discrete random variable can take ………… number of values with in its range
	(g) The range of distribution function is ……..
	(h) If X and Y are two random variables such that their expectation exist and $P(x \leq y) = 1$ then (i) $E(X) \leq E(Y)$ (ii) $E(X) = E(Y)$ (iii) $E(X) \geq E(Y)$ (iv) None of these
	(i) The height of persons in a country is a random variable of the type ………

SECTION B (Concept Based Questions)

Q2. Two dice are rolled. Let X denote the random variable which counts the total number of points on the upturned faces, construct a table giving the non-zero values of the probability mass function. Also find the distribution function of X.

Ans. F(x)= $\begin{cases} \mathbf{0}, & \boldsymbol{x < 2} \\ \frac{1}{36}, & \boldsymbol{2 \leq x < 3} \\ \frac{3}{36}, & \boldsymbol{3 \leq x < 4} \\ \frac{6}{36}, & \boldsymbol{4 \leq x < 5} \\ \frac{35}{35}, & \boldsymbol{11 \leq x < 12} \\ \mathbf{1}, & \boldsymbol{x \geq 12} \end{cases}$

Q3. A diameter of an electric cable, say X, is assumed to be a continuous random variable with p.d.f. : $f(x) = 6x(1-x),\ 0 \leq x \leq 1$.
Check that $f(x)$ is p.d.f., and (ii) determine a number b such that $P(X < b) = P(X > b)$. (iii) Obtain expression for the c.d.f. of X. (iv) Compute $P\left(X \leq \frac{1}{2} \middle| \frac{1}{3} \leq X \leq \frac{2}{3}\right)$.

Ans. ii. b=1/2 **iii.** $\boldsymbol{F(x) = 3x^2 - 2x^3, 0 \leq x \leq 1,}$ **iv. 11/26**

Q4. Let X be a random variable with the following probability distribution :

x	-3	6	9
$P(X = x)$	1/6	1/2	1/3

	Find $E(X)$ and $E(X^2)$ and using the laws of expectation, evaluate $E(2X+1)^2$. **Ans. E(X)=11/2, $E(X^2)=93/2$, 209**
Q5.	If the Chebyshev's inequality for the random variable X is given by $P(-2<X<8)\geq\frac{21}{25}$, find $E(X)$ and $Var(X)$. **Ans. 3, 4**
	SECTION-C (Analytical Based Questions)
Q6.	A random variable X has the following probability function:

Values of X, x:	0	1	2	3	4	5	6	7
P(x)	0	k	2k	2k	3k	K^2	$2k^2$	$7k^2$+k

(i) Find k, (ii) Evaluate $P(X<6), P(X\geq 6)$, and $P(0<X<5)$ (iii) determine the distribution function of X (iv) if $P(X\leq a)>\frac{1}{2}$, find the minimum value of a.

Ans. (i) k=1/10, ii. 81/100, 19/100 , 4/5 iv. A=4

Q7.	A random number X is distributed at random between the values 0 and 1 so that its probability density function is $f(x)=kx^2(1-x^3)$, where k is a constant. Find the value of k. Using this value of k, find its mean and variance. **Ans. 6, 9/14, 9/245**
Q8.	The distribution function of the random variable X is given by $F(x)=\begin{cases}0, & x\leq 0\\ \frac{(x-1)^4}{16}, & 1\leq x\leq 3\\ 1, & x>3\end{cases}$ i. Find corresponding density function. ii. Compute P($2\leq x\leq 3$) **Ans. ii. 15/16**
Q9.	If X and Y are independent random variables with mean 2, 3 and variance 1, 2 respectively. Find the the mean and variance of the random variable $Z=2X-5Y$. **Ans. -11, 54**
Q10.	A symmetric die is thrown 600 times. Find the lower bound for the probability of getting 80 to 120 sixes. **Ans. 19/24**
	SECTION-D (Application based Question)
Q11.	The distribution function of continuous random variable X given by $F(x)=\begin{cases}0, & x<0\\ \frac{x}{2}, & 0\leq x<1\\ \frac{1}{2}, & 1\leq x<2\\ \frac{x}{4}, & 2\leq x<4\\ 1, & x\geq 4\end{cases}$ Find probability density function of X.
Q12.	Suppose that the life in hours of a certain part of radio tube is a continuous random variable X with p.d.f. given by

	$$f(x)=\begin{cases}\frac{100}{x^2}, & when\ x\geq 100\\ 0, & elsewhere\end{cases}$$ (i) What is the probability that all of three such tubes in a given radio set will have to be replaced during the first 150 hours of operation? (ii) What is probability that none of three of the original tubes will have to be replaced during that first 150 hours of operation? (iii) What is probability that a tube will last less than 200 hours if it is known that the tube is still functioning after 150 hours of service? (iv) What is the maximum number of tubes that may be inserted into a set so that there is a probability of 0.5 that after 150 hours of service all of them are still functioning? **Ans. 1/27, 2/3, 0.25, 1.7**
Q13.	For probability distribution function $p(x)=2^{-x}; x=1,2,3\ldots$, prove that chebyschev's inequality gives $P\{\lvert X-2\rvert\leq 2\}>\frac{1}{2}$, while the actual probability is $\frac{15}{16}$.
Q14.	A random variable x has the pdf $$f(x)=\begin{cases}ax, & 0\leq x<1\\ a, & 1\leq x<2\\ -ax+3a, & 2\leq x<3\\ 0, & Otherwise\end{cases}$$ Find i. Value of a ii. P(X$\leq$ 5) **Ans. i. a=1/2, ii. 1/2**
Q15.	Let $Y=X^2+2X$, where X is a random variable whose probability distribution is given as: <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>P(x)</td><td>0.1</td><td>0.2</td><td>0.5</td><td>0.1</td></tr></table>Then find probability distribution, CDF, mean, and variance of Y. **Ans. Mean=11.8,**